

CYBER SECURITY INTERNSHIP WEEK 1 ASSIGNMENT REPORT

Submitted By:

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Internship Organisation:

DG Interns Hub

SIN: DG/SEPTEMBER/CYBER/130

Topic:

Networking Fundamentals

Week:

Week 1

Submission Date:

7 September 2026

Task 1. Introduction to OSI Model

The OSI (Open Systems Interconnection) Model is a conceptual model that explains how data is communicated between devices over a network. It divides network communication into seven layers, and each layer performs a specific function.

1. Physical Layer — Layer 1

Function: Transmits raw bits (0s and 1s) through physical media. It deals with cables, signals, connectors, and physical network equipment. Eg's Ethernet cables, fibre-optic cables, radio signals, and hubs.

Real-life example: When data travels through an Ethernet cable, this layer handles the physical signals carrying the data.

2. Data Link Layer — Layer 2

Function: Provides communication between devices on the same local network. It organises data into frames and uses MAC addresses to identify devices. Eg's Ethernet, Wi-Fi, MAC addresses, and switches.

Real-life example: A switch uses MAC addresses to forward data to the correct device on a local network.

3. Network Layer — Layer 3

Function: Handles IP addressing and routing. It determines how packets travel from the source to the destination across different networks. Eg's IPv4, IPv6, IP addresses, and routers.

Real-life example: When you visit a website, routers use IP addresses to forward your packets toward the web server.

4. Transport Layer — Layer 4

Function: Provides end-to-end communication between applications. It can provide reliable delivery, error recovery, and proper sequencing of data. Eg's TCP and UDP.

Real-life example: TCP helps ensure that data from a file download arrives reliably and in the correct order.

5. Session Layer — Layer 5

Function: Establishes, manages and terminates communication sessions between applications. Eg's Session establishment, session management, and session termination.

Real-life example: When two applications communicate, this layer helps maintain the communication session until it is finished.

6. Presentation Layer — Layer 6

Function: Handles the format and representation of data. It can also perform encryption, decryption, and compression. Eg's Data formatting, encryption, compression, and character encoding.

Real-life example: Data may be encrypted or converted into a suitable format before being sent across a network.

7. Application Layer — Layer 7

Function: Provides network services directly to applications and is the layer closest to the user. Eg's HTTP, HTTPS, FTP, DNS, and SMTP.

Real-life example: When you open a website in Safari, HTTPS is used for communication between the browser and web server.

OSI Model Summary

Layer	Name	Main Function
7	Application	Network services for applications
6	Presentation	Data formatting and encryption
5	Session	Manages communication sessions
4	Transport	End-to-end communication
3	Network	IP addressing and routing
2	Data Link	Frames and MAC addressing
1	Physical	Transmission of bits

TASK 2: TCP/IP MODEL

1. Introduction: The TCP/IP (Transmission Control Protocol/Internet Protocol) Model is a networking model used for communication over the Internet and other computer networks. It describes how data is transmitted between devices by dividing the communication process into four layers.

The four layers are:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Access Layer

2. Application Layer

Function: The Application Layer is the top layer of the TCP/IP Model. It provides network services to applications and allows users to access network resources and services. Eg's

- HTTP/HTTPS
- FTP
- DNS
- SMTP

Real-Life Example: When you open a website in a web browser, the Application Layer uses protocols such as HTTPS to communicate with the web server.

3. Transport Layer

Function: The Transport Layer provides communication between applications running on different devices. It manages data delivery and can provide reliability, error control, and flow control. Eg's

- TCP
- UDP

Real-Life Example: When downloading a file, TCP helps ensure that the data reaches your device reliably and in the correct order.

4. Internet Layer

Function: The Internet Layer is responsible for logical addressing and routing packets between different networks. It uses IP addresses to identify the source and destination of data. Eg's

- IPv4
- IPv6
- ICMP

Real-Life Example: When you send a request to a website, routers use IP addresses to forward packets toward the destination server.

5. Network Access Layer

Function: The Network Access Layer is responsible for transferring data over the physical network. It handles communication between devices on the local network and includes functions related to frames, MAC addresses, and physical transmission. Eg's

- Ethernet
- Wi-Fi
- MAC addresses
- Network cables

Real-Life Example: When your laptop connects to your Wi-Fi router, this layer handles the local network communication needed to transmit data.

6. OSI Model vs TCP/IP Model

Feature	OSI Models	TCP/IP Model
Number of layers	7	4
Type	Reference/conceptual model	Practical networking model
Application	Application, Presentation, Session	Application
Transport	Transport	Transport
Network	Network	Internet
Data Link + Physical	Link and Physical	Network Access
Main purpose	Helps understand network communication	Used for real-world Internet

TASK 3: NETWORKING PROTOCOLS

1. Introduction: A network protocol is a set of rules that determine how devices communicate and exchange data over a network. Different protocols perform different functions, such as accessing websites, transferring files, finding IP addresses, and automatically configuring network devices. The important protocols studied in this task are HTTP/HTTPS, FTP, DNS, and DHCP.

2. HTTP and HTTPS

What it does: HTTP (Hypertext Transfer Protocol) is used to transfer information between a web browser and a web server. It is commonly used for accessing websites and transferring web pages and other web resources. HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP. It uses encryption through TLS to help protect data exchanged between the browser and the web server.

Where it is used: HTTP and HTTPS are mainly used for:

- Websites
- Web applications
- Online services
- Communication between browsers and web servers

Real-Life Example: When you open a website in Safari or another browser, the browser communicates with the website server using HTTP or, more commonly today, HTTPS.

3. FTP

What it does: FTP (File Transfer Protocol) is a protocol used to transfer files between a client and a server over a network. It allows users to upload files to a server and download files from it.

Where it is used: FTP can be used for:

- Uploading files to servers
- Downloading files
- Managing files on remote servers
- Transferring files within organizations

Real-Life Example: A website administrator can use an FTP client to upload website files from a computer to a web server.

4. DNS

What it does: DNS (Domain Name System) translates human-readable domain names into IP addresses. This allows users to access websites using names instead of remembering numerical IP addresses.

example: google.com → IP address of Google's server

Where it is used: DNS is used whenever a device needs to find the IP address associated with a domain name.

Real-Life Example: When you type a website name into your browser, DNS helps find the server's IP address, and your browser can connect to it.

5. DHCP

What it does: DHCP (Dynamic Host Configuration Protocol) automatically provides network configuration information to devices. This can include an IP address, subnet mask, default gateway, and DNS server information.

Where it is used: DHCP is commonly used in:

- Home networks
- Offices
- Schools
- Public Wi-Fi networks

Real-Life Example: When you connect your Mac or phone to your home Wi-Fi, your router can use DHCP to automatically assign your device to an IP address.

6. Protocol Summary

Protocol	Full Form	Main Purpose
HTTP	Hypertext Transfer Protocol	Transfers web information
HTTPS	Hypertext Transfer Protocol Secure	Secure web communication
FTP	File Transfer Protocol	Transfers files
DNS	Domain Name System	Converts domain names to addresses
DHCP	Dynamic Host Configuration Protocol	Automatically assigns network configuration

TASK 4: IP ADDRESSING

1. What is an IP Address?

An IP (Internet Protocol) address is a unique logical address assigned to a device connected to a network. It helps identify the device and allows data to be delivered to the correct destination.

An IP address is similar to an address used for delivering a letter. Just as a postal address helps deliver a letter to the correct location, an IP address helps network devices identify where data should be sent.

Example: An IPv4 address can look like: 192.168.1.10

An IPv4 address consists of four numbers separated by dots, with each number ranging from 0 to 255.

2. Public and Private IP Addresses

IP addresses can be broadly categorised as public and private addresses.

Public IP Address: A public IP address is an address used to identify a network or device on the Internet. It is normally assigned to a network by an Internet Service Provider (ISP).

Where it is used:

- Internet communication
- Websites and servers
- Home and office Internet connections

Example: Your home router may have a public IP address provided by your ISP.

Private IP Address: A private IP address is used within a local network, such as a home, school, or office network. Devices using private IP addresses communicate with other devices inside the local network.

Common private IPv4 ranges include:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

Example: Your computer might have: 192.168.1.10

while your router might use: 192.168.1.1

3. Public IP vs Private IP

Feature	Public IP	Private IP
Used for	Internet communication	Local network communication
Assigned by	Usually ISP	Router/DHCP or administrator
Internet accessible	Yes when appropriately configured	Not directly routable on the public Internet
Example	ISP-provided address	192.168.1.10

4. Basic Subnetting Concept: Subnetting is the process of dividing a larger network into smaller networks called subnets. A subnet mask determines which portion of an IP address represents the network, and which portion represents the host.

Example: Consider: IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0

The subnet mask 255.255.255.0 is commonly written as /24.

In a typical 192.168.1.0/24 network:

- Network address: 192.168.1.0
- Usable host addresses: 192.168.1.1 – 192.168.1.254
- Broadcast address: 192.168.1.255

This means the network can provide 254 usable IPv4 host addresses.

5. Default Gateway: A default gateway is the device, usually a router, that forwards traffic from a local network to other networks.

Example: If your computer has: IP: 192.168.1.10
Subnet Mask: 255.255.255.0
Gateway: 192.168.1.1

Then, 192.168.1.1 is the router/default gateway used to reach other networks.

Real-Life Example

When your Mac is connected to your home Wi-Fi and you open a website, your computer sends traffic through the router, which acts as the default gateway to reach the Internet.

TASK 5: NETWORK COMMANDS

1. Introduction

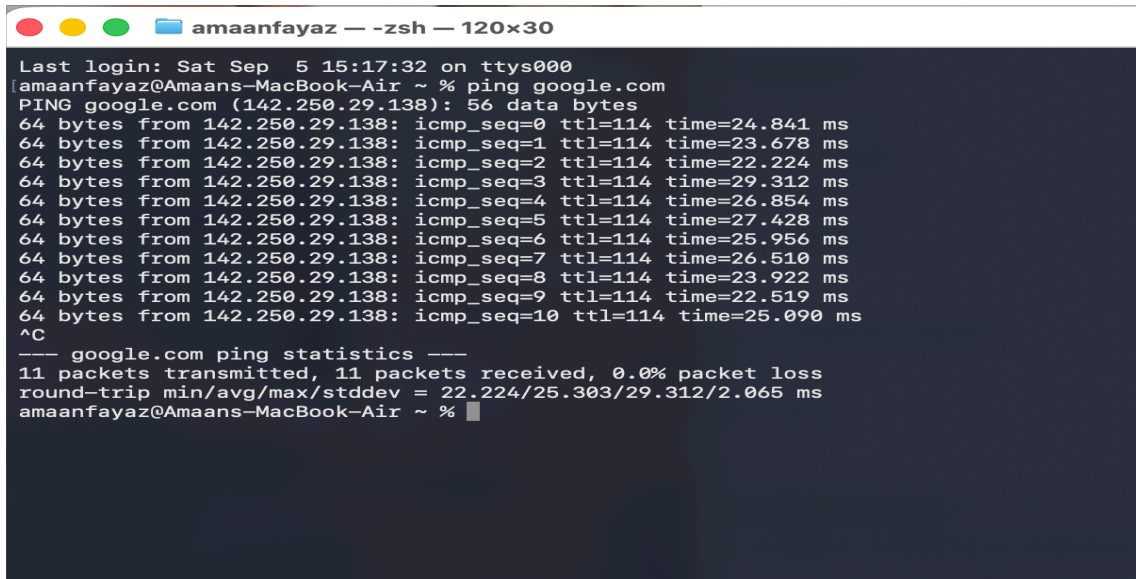
Network commands are useful tools for checking network connectivity, viewing network configuration, and analysing the path taken by data between devices. In this task, three commands were performed on macOS: ping, ifconfig, and trace route.

2. Ping Command

Command: ping google.com

Purpose: The ping command checks whether a destination is reachable over the network. It sends packets to the destination and measures the response time.

Result: The test successfully received 11 packets with 0.0% packet loss, showing that the connection to Google was working successfully.



```
amaanfayaz — zsh — 120x30
Last login: Sat Sep  5 15:17:32 on ttys000
amaanfayaz@Amaans-MacBook-Air ~ % ping google.com
PING google.com (142.250.29.138): 56 data bytes
64 bytes from 142.250.29.138: icmp_seq=0 ttl=114 time=24.841 ms
64 bytes from 142.250.29.138: icmp_seq=1 ttl=114 time=23.678 ms
64 bytes from 142.250.29.138: icmp_seq=2 ttl=114 time=22.224 ms
64 bytes from 142.250.29.138: icmp_seq=3 ttl=114 time=29.312 ms
64 bytes from 142.250.29.138: icmp_seq=4 ttl=114 time=26.854 ms
64 bytes from 142.250.29.138: icmp_seq=5 ttl=114 time=27.428 ms
64 bytes from 142.250.29.138: icmp_seq=6 ttl=114 time=25.956 ms
64 bytes from 142.250.29.138: icmp_seq=7 ttl=114 time=26.510 ms
64 bytes from 142.250.29.138: icmp_seq=8 ttl=114 time=23.922 ms
64 bytes from 142.250.29.138: icmp_seq=9 ttl=114 time=22.519 ms
64 bytes from 142.250.29.138: icmp_seq=10 ttl=114 time=25.090 ms
^C
--- google.com ping statistics ---
11 packets transmitted, 11 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 22.224/25.303/29.312/2.065 ms
amaanfayaz@Amaans-MacBook-Air ~ %
```

3. ifconfig Command

Command: ifconfig

Purpose: The ifconfig command displays information about network interfaces, including IP addresses, MAC addresses, interface status, and other network configuration details.

Result: The command displayed the network interfaces and their configuration on my Mac.

```
amaanfayaz@Amaans-MacBook-Air ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=1208<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
    inet 127.0.0.1 netmask 0xffff0000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anp10: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether b6:a6:dc:9a:2d:0a
    media: none
    status: inactive
anp11: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether b6:a6:dc:9a:2d:0b
    media: none
    status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether b6:a6:dc:9a:2d:ea
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether b6:a6:dc:9a:2d:eb
```

4. Trace route Command

Command: trace route google.com

Purpose: Trace route shows the path taken by network packets from the computer to the destination. Each numbered entry represents a network hop.

Result: The command displayed multiple hops between my Mac and Google's server. Some hops showed * * *, meaning there was no response from those hops.

```
nd6 options=201<PERFORMNUD,DAD>
[amaanfayaz@Amaans-MacBook-Air ~ % traceroute google.com
traceroute: Warning: google.com has multiple addresses; using 142.250.122.138
traceroute to google.com (142.250.122.138), 64 hops max, 40 byte packets
 1 dsldevice.lan (192.168.1.1)  4.470 ms  3.427 ms  3.159 ms
 2 10.240.15.104 (10.240.15.104)  6.633 ms  6.525 ms  9.858 ms
 3 152.52.51.213 (152.52.51.213)  8.666 ms  10.052 ms
 4 152.52.51.85 (152.52.51.85)  7.786 ms
 5 116.119.42.201 (116.119.42.201)  21.988 ms  22.144 ms  18.669 ms
 6 142.250.161.56 (142.250.161.56)  22.042 ms  21.304 ms  22.828 ms
 7 142.251.76.107 (142.251.76.107)  24.044 ms  22.463 ms
 8 142.251.193.131 (142.251.193.131)  27.393 ms
 9 216.239.62.180 (216.239.62.180)  24.726 ms  24.363 ms  23.942 ms
10 192.178.83.95 (192.178.83.95)  23.143 ms
11 142.251.226.81 (142.251.226.81)  24.126 ms
12 142.251.66.229 (142.251.66.229)  24.511 ms
13 142.250.239.106 (142.250.239.106)  20.727 ms
14 142.250.239.104 (142.250.239.104)  24.133 ms
15 142.250.208.70 (142.250.208.70)  23.391 ms
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TASK 6: NETWORK SIMULATION USING CISCO PACKET TRACER

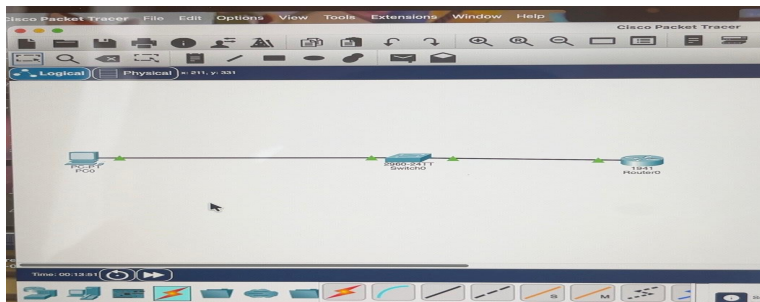
1. Introduction:

Cisco Packet Tracer is a network simulation tool used to create and test computer networks virtually. In this task, I created a simple network consisting of a PC, a switch, and a router. I configured IP addresses and tested connectivity between the PC and the router.

1. Network Topology: The network was created using the following devices:

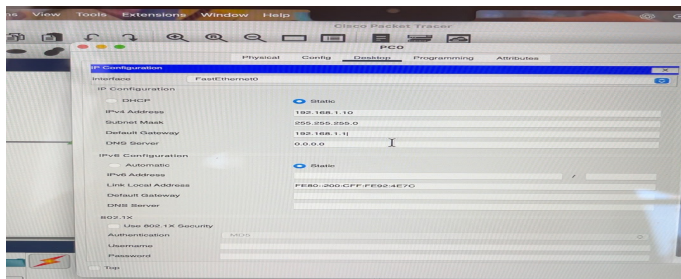
- PC0 – End device
- 2960-24TT Switch0 – Network switch
- 1941 Router0 – Router
- Copper Straight-Through Cable – Used to connect the devices

2. Topology: PC0 → Switch0 → Router0



3. IP Address Configuration

Device.	IP Address	Sub Net Mask	Default
Gateway			
PC0	192.168.1.10	255.255.255.0	192.168.1.1
Router0	192.168.1.1	255.255.255.0	—



4. Connectivity Testing

After configuring the devices, I tested the connection from PC0 using ping 192.168.1.1

The test was successful. 4 packets were sent, 4 packets were received, and there was a 0% packet loss. This confirmed that PC0 could successfully communicate with Router0.

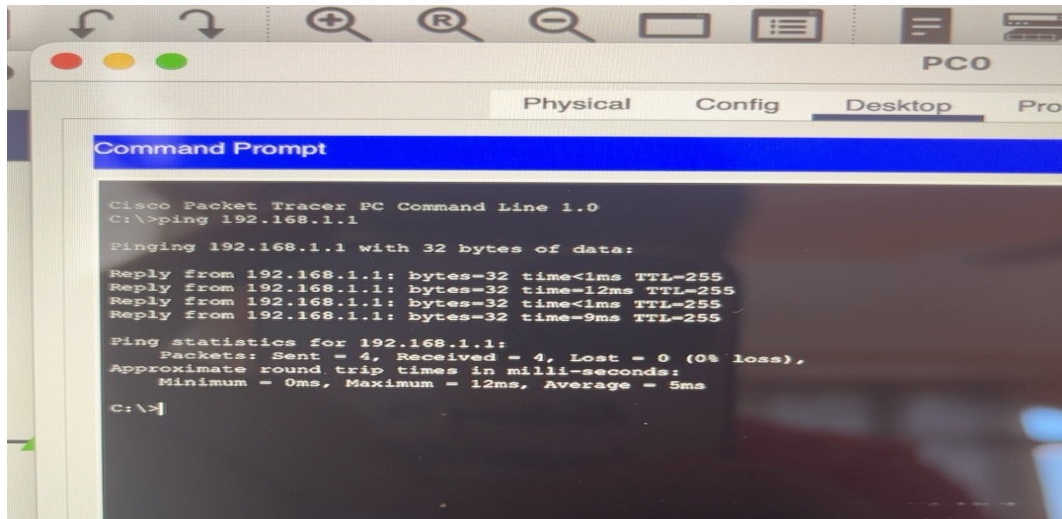


Figure this shows successful connectivity test between PC0 and Router0

5. Result

The network simulation was successfully completed. The PC, switch, and router were connected using Copper Straight-Through cables, the required IP addresses were configured, and connectivity was verified using the ping command.

6. Conclusion

This task helped me understand how basic network devices are connected and configured. I learned how to assign IP addresses, enable a router interface, connect devices, and test network connectivity using Cisco Packet Tracer.